Rajat Sharma

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Summary

Backend and platform engineer with **5+ years** building microservices, event-driven systems and multi-tenant SaaS with **Java**, **Go**, **Spring Boot**, **Kafka**, **Cassandra** and **Kubernetes**, currently at **IBM**. I also **build with AI**: native apps that run speech and language models **on device**.

Technical Skills

Languages: Java, Go, Python, TypeScript, SQL, Scala, C++, Bash

Backend & Data: Spring Boot, FastAPI, Kafka, Redis Streams, Cassandra, PostgreSQL, Db2, MongoDB, Redis

Cloud, DevOps & Observability: Kubernetes, OpenShift, Docker, Helm, Tekton, Argo CD, Terraform, AWS, Grafana, Instana, JUnit, JMH

Experience

IBM - Software Developer, Maximo Small SaaS — Markham, Ontario
Go, Python, Kubernetes, OpenShift, Helm, Tekton, Argo CD, Terraform, AWS, Db2, Next.js — *April 2026 – Present*

- Build a **multi-tenant Maximo Manage-as-a-Service** platform on **Kubernetes/OpenShift** across **19 container images** and **11 Helm charts**, with every change gated by Helm tests and **Tekton** pipelines.
- Re-keyed tenant identifiers to **enable tenant migration** between tenant groups, and automated **Amazon RDS Db2** setup with stored procedures, removing manual provisioning steps.
- Built a **Go tenant-health API** and a **Next.js SRE dashboard** showing provisioning and stuck tenants.
- Closed **SQL-injection** paths and Go CVEs, added secret scanning, and wrote **Terraform** for AWS accounts.

IBM - Software Developer, Envizi — Markham, Ontario
Java, Spring Boot, Python, Cassandra, Kafka, Redis, Kubernetes — *October 2024 – March 2026*

- Designed a **denormalized Cassandra schema** (**4 query-specific tables**) for hierarchical organization data, cutting lookup latency by **65%** across **4 production clusters**.
- Built a **Kafka monitoring service** for **30+ topics** and **500K+ events/day** with Redis-backed locking and alert deduplication, cutting alert noise by **90%** with sub-minute outage detection.
- Cut XBRL taxonomy loads from **5s to 1s** with LRU caching; patched **Kafka CVE-2025-27817** in **5+ services**.

Concordia University - Developer — Montreal, Quebec
Java, Spring Boot, Angular, MongoDB, Kubernetes — *September 2022 – January 2024*

- Cut deployment time and developer workload by **44%**; built **Spring Boot APIs** used by **12+ applications**.

Tejas Networks - Software Developer — New Delhi, India
Java, Spring Boot, Kafka, SQL — *August 2021 – August 2022*

- Modernized **TejNMS** with Spring Boot (**30%** better resource allocation); led **3** engineers moving **ActiveMQ to Kafka** (**20%** faster).

Projects

Utter | AI voice layer for macOS | *Swift, SwiftUI, Whisper, MLX, Apple Foundation Models* — *November 2026*

- Talk and polished text lands in any app: **Whisper on Apple MLX** transcribes **on device** (**0.6 s** for a 7 s dictation) and **Apple's on-device model** cleans it up. No audio leaves the Mac.

Kaleido TV | **GitHub** | *Swift, SwiftUI, AVFoundation, HLS, XMLTV, Keychain* — *October 2026*

- Live TV for **Mac, iPhone and iPad**: a checker loads real video to rank working channels; playback **reconnects with backoff** and fails over to backup streams. No third-party packages.

LSM-KV Storage Engine | **GitHub** | *Java 21, Gradle, JUnit, JMH, Docker, Kubernetes* — *October 2026*

- LSM-tree store from scratch: CRC-checked **write-ahead log**, memory-mapped **SSTables** with **Bloom filters**, **k-way merge** compaction. **3 μs** writes (JMH); missing-key lookups **17x** cheaper.

Loonie Ledger | *Java 21, Spring Boot, Python, TypeScript, Redis Streams, PostgreSQL, Kubernetes* — *October 2026*

- Neobank of **3 event-driven services in 3 languages**: **double-entry ledger** with a **transactional outbox**, fraud and compliance on **Redis Streams**, and a **SHA-256 hash-chained** audit log.

Education & Achievements

Master of Applied Computer Science, Concordia University; GPA 3.6/4.3 — *Montreal | 2022 – 2024*
B.Tech, IT, Guru Gobind Singh Indraprastha University; GPA 3.9/4.3 — *New Delhi | 2017 – 2021*
LeetCode Knight (2018), CodeChef 4 Star (1848), HackerEarth (1676); **Golden Key Scholar**.